

Math 521
Homework 4

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Problem 1. Prove Cauchy's inequality with $\epsilon > 0$: If $a, b \in \mathbb{R}$ then:

$$ab \leq \frac{\epsilon a^2}{2} + \frac{b^2}{2\epsilon}.$$

Proof. Since $a, b \in \mathbb{R}$, we have:

$$0 \leq (a - b)^2.$$

Expanding the right-hand side and isolating the ab term yields:

$$ab \leq \frac{a^2}{2} + \frac{b^2}{2}.$$

Now let $\epsilon > 0$. Since this holds true for all $a', b' \in \mathbb{R}$, take $a' = \sqrt{\epsilon}a$ and $b' = \frac{b}{\sqrt{\epsilon}}$. Then:

$$\begin{aligned} ab &= \sqrt{\epsilon}a \cdot \frac{b}{\sqrt{\epsilon}} \\ &\leq \frac{(\sqrt{\epsilon}a)^2}{2} + \frac{\left(\frac{b}{\sqrt{\epsilon}}\right)^2}{2} \\ &= \frac{\epsilon a^2}{2} + \frac{b^2}{2\epsilon}. \end{aligned}$$

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Problem 2. Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1 . Let $u \in C^2(\overline{U})$ with $u = 0$ on ∂U . Prove the following interpolation inequality:

$$\int_U |Du|^2 dx \leq \frac{\epsilon}{2} \int_U (\Delta u)^2 dx + \frac{1}{2\epsilon} \int_U u^2 dx$$

Proof. From the previous problem, take $a = u(x)$ and $b = \Delta u(x)$. Then:

$$-u(x)\Delta u(x) \leq \frac{\epsilon(\Delta u(x))^2}{2} + \frac{(-u(x))^2}{2\epsilon}.$$

Integrating both sides over U gives:

$$-\int_U u \Delta u dx \leq \frac{\epsilon}{2} \int_U (\Delta u)^2 dx + \frac{1}{2\epsilon} \int_U u^2 dx$$

Since $u = 0$ on ∂U , we can rewrite the left-hand side of the inequality as:

$$-\int_U u \Delta u dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u dS \leq \frac{\epsilon}{2} \int_U (\Delta u)^2 dx + \frac{1}{2\epsilon} \int_U u^2 dx.$$

Green's identity yields:

$$\int_U Du \cdot Du dx \leq \frac{\epsilon}{2} \int_U (\Delta u)^2 dx + \frac{1}{2\epsilon} \int_U u^2 dx.$$

Rewriting $Du \cdot Du = |Du|^2$ gives the desired inequality.

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Problem 3. Assuming f approaches zero rapidly enough as $x \rightarrow \pm\infty$, find a function u which satisfies the ODE problem:

$$u''(x) - u(x) = f(x).$$

Your solution should be of the form:

$$u(x) = \int_{-\infty}^{\infty} K(x, t) f(x - t) dt.$$

Proof. After taking the Fourier transform of both sides, our ODE becomes:

$$(iy)^2 \hat{u} - \hat{u} = \hat{f}.$$

Solving for $\hat{u}$ gives:

$$\hat{u}(y) = \frac{-1}{1 + y^2} \hat{f}.$$

Define $h(y) = \frac{-1}{1+y^2}$. Taking the inverse Fourier transform of both sides yields:

$$\begin{aligned} u(x) &= \mathcal{F}^{-1}(h(y)\hat{f}(y)) \\ &= (\check{h} * f)(x) \\ &= \int_{-\infty}^{\infty} \check{h}(t) f(x - t) dt. \end{aligned}$$

We previously showed that $\mathcal{F}(e^{-a|x|}) = \frac{1}{\sqrt{2\pi}} \frac{2a}{a^2 + y^2}$. Hence:

$$\check{h}(y) = \mathcal{F}^{-1}\left(\frac{-1}{1 + y^2}\right) = -\frac{\sqrt{2\pi}}{2} e^{-|x|}.$$

Thus the solution to our ODE is:

$$u(x) = -\frac{\sqrt{2\pi}}{2} \int_{-\infty}^{\infty} e^{-|t|} f(x - t) dt. \quad \blacksquare$$

Problem 4. For $x \in \mathbb{R}$, define $v(x) := e^{-\frac{1}{2}x^2}$. Show that $\hat{v}(y) = e^{-\frac{1}{2}y^2}$.

Problem 5. Let $a > 0$ and $f \in L^1(\mathbb{R})$. Define $g(x) := f(ax)$. Show that:

$$\hat{g}(y) = \frac{1}{a} \hat{f}\left(\frac{y}{a}\right).$$

Proof. Observe that:

$$\begin{aligned} \hat{g}(y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-ixy} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(ax) e^{-ixy} dx. \end{aligned}$$

The substitution $x = \frac{t}{a}$ gives $dx = \frac{1}{a} dt$. Hence:

$$\begin{aligned} \hat{g}(y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{a} f(t) e^{-it \frac{y}{a}} dy \\ &= \frac{1}{a} \hat{f}\left(\frac{y}{a}\right). \end{aligned} \quad \blacksquare$$

Problem 6. For $x \in \mathbb{R}$ and $\epsilon > 0$, let $v_\epsilon(x) := e^{-\epsilon x^2}$. Use the previous two problems to show that:

$$\widehat{v}_\epsilon(y) = \frac{1}{\sqrt{2\epsilon}} e^{-\frac{y^2}{4\epsilon}}.$$

Proof. Let $f(x) = e^{-\frac{1}{2}x^2}$. Then $v_\epsilon(x) = f(\sqrt{2\epsilon}x)$. By Problem 4 and 5, we have:

$$\begin{aligned} \widehat{v}_\epsilon(y) &= \mathcal{F}\left(f\left(\sqrt{2\epsilon}x\right)\right) \\ &= \frac{1}{\sqrt{2\epsilon}} \widehat{f}\left(\frac{y}{\sqrt{2\epsilon}}\right) \\ &= \frac{1}{\sqrt{2\epsilon}} f\left(\frac{y}{\sqrt{2\epsilon}}\right) \\ &= \frac{1}{\sqrt{2\epsilon}} e^{-\frac{y^2}{4\epsilon}}. \end{aligned} \quad \blacksquare$$

Problem 7. For $x \in \mathbb{R}^n$ and $\epsilon > 0$, let $v_\epsilon(x) := e^{-\epsilon |x|^2}$. Use Fubini's theorem and the previous problem to show that:

$$\widehat{v}_\epsilon(y) = \frac{1}{(2\epsilon)^{\frac{n}{2}}} e^{-\frac{|y|^2}{4\epsilon}}.$$

Proof. First note that we can rewrite $v_\epsilon(x)$ and $e^{-ix \cdot y}$ as:

$$\begin{aligned} v_\epsilon(x) &= e^{-\epsilon |x|^2} = \prod_{j=1}^n e^{-\epsilon x_j^2}, \\ e^{-ix \cdot y} &= \prod_{j=1}^n e^{-ix_j y_j}. \end{aligned}$$

Using this fact and Fubini's theorem yields:

$$\begin{aligned} \widehat{v}_\epsilon(y) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} v_\epsilon(x) e^{-ix \cdot y} dx \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^n \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{j=1}^n e^{-\epsilon x_j^2} e^{-ix_j y_j} dx_1 \dots dx_n. \end{aligned}$$

For a fixed j , note that $\int_{-\infty}^{\infty} e^{x_j^2} e^{-ix_j y_j} dx_j$ is viewed as a constant when integrated against any other x_k such that $k \neq j$. Using this fact, and by Problem 6, we have:

$$\begin{aligned}\widehat{v}_\epsilon &= \left(\frac{1}{\sqrt{2\pi}} \right)^n \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{j=1}^n e^{-x_j^2} e^{-ix_j y_j} dx_1 \dots dx_n \\ &= \prod_{j=1}^n \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x_j^2} e^{-ix_j y_j} dx_j \\ &= \prod_{j=1}^n \frac{1}{\sqrt{2\epsilon}} e^{-\frac{y_j^2}{4\epsilon}} \\ &= \frac{1}{(2\epsilon)^{\frac{n}{2}}} e^{-\frac{|y|^2}{4\epsilon}}.\end{aligned}$$

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Problem 8. Show that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. Then use this to show that:

$$\int_{\mathbb{R}^n} e^{-|x|^2} dx = \pi^{\frac{n}{2}}.$$

Proof. Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$. Then:

$$\begin{aligned}I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 \\ &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{2} \int_0^{\infty} e^{-u} du \right) d\theta \\ &= \int_0^{2\pi} \frac{1}{2} d\theta \\ &= \pi.\end{aligned}$$

Thus $I = \sqrt{\pi}$. Hence:

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-|x|^2} dx &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{j=1}^n e^{-x_j^2} dx_1 \dots dx_n \\ &= \prod_{j=1}^n \int_{-\infty}^{\infty} e^{-x_j^2} dx_j \\ &= \prod_{j=1}^n \sqrt{\pi} \\ &= \pi^{\frac{n}{2}}.\end{aligned}$$

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